

NAME: N.ROHITH

ID NO: 2520090191

PRACTICAL WEEK 3

Task 1: Develop a C program using fork() that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process States at different stages of execution

CODE:

```
UW PID0 6.09 File: lk.c

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;

    printf("=== Process Creation using fork() ===\n");
    printf("Before fork()\n");
    printf("PID : %d\n", getpid());
    printf("PPID : %d\n", getppid());
    printf("State: Running\n");

    pid = fork();

    if (pid < 0)
    {
        printf("Fork failed.\n");
        return 1;
    }

    if (pid == 0)
    {
        /* Child Process */
        printf("=== Child Process ===\n");
        printf("PID : %d\n", getpid());
        printf("PPID : %d\n", getppid());
        printf("State: Running\n");

        sleep(2);

        printf("\nChild Process:\n");
        printf("PID : %d\n", getpid());
        printf("PPID : %d\n", getppid());
        printf("State: Exiting\n");

        exit(0);
    }
    else
    {
        /* Parent Process */
        printf("=== Parent Process ===\n");
        printf("PID : %d\n", getpid());
        printf("PPID : %d\n", getppid());
        printf("Child PID: %d\n", pid);
        printf("State: Running\n");

        printf("\nParent is waiting for child...\n");
        printf("State: Waiting\n");

        wait(NULL);

        printf("\nChild process has terminated.\n");
        printf("Parent Process:\n");
        printf("PID : %d\n", getpid());
        printf("PPID : %d\n", getppid());
    }
}
```

OUTPUT:

```
nrk@nallas-mac-Book-Air ~ % gcc lk.c -o lk
nrk@nallas-mac-Book-Air ~ % ./lk
=== Process Creation using fork() ===
Before fork():
PID : 1541
PPID : 1541
State: Running

=== Parent Process ===
PID : 1541
PPID : 1541
Child PID: 1562
State: Running

Parent is waiting for child...
State: Waiting
=== Child Process ===
PID : 1562
PPID : 1561
State: Running

Child Process:
PID : 1562
PPID : 1561
State: Exiting

Child process has terminated.
Parent Process:
PID : 1541
PPID : 1541
State: Running

Parent Process State: Exiting
nrk@nallas-mac-Book-Air ~ %
```

Task-2: Using Linux terminal commands (uname, lscpu, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

Code:

```
UW PIDO 6.09 File: 13.c

#include <stdio.h>
#include <stdlib.h>

int main()
{
    printf("=====\\n");
    printf("  HARDWARE AND OS RESOURCE ANALYSIS\\n");
    printf("=====\\n\\n");

    printf("1. OPERATING SYSTEM INFORMATION\\n");
    printf("-----\\n");
    system("uname -a");

    printf("2. CPU INFORMATION\\n");
    printf("-----\\n");
    system("lscpu");

    printf("3. RUNNING PROCESSES\\n");
    printf("-----\\n");
    system("ps aux");

    printf("4. REAL-TIME SYSTEM RESOURCE USAGE\\n");
    printf("-----\\n");
    printf("Press Ctrl+C to exit the top command.\\n\\n");
    system("top");

    return 0;
}
```

Output:

Processes: 633 total, 2 running, 631 sleeping, 3217 threads.																										08:37:09	
Load Avg: 3.68, 110.26, 145.48 CPU usage: 4.43% user, 5.90% sys, 89.63% idle. Sharedlib: 358M resident, 80M data, 64M linkedin. MemRegions: 0 total, 88 resident, 185M private, 787M shared.																											
PhysMem: 7444M used (1024M wired, 7837M compressed), 140M unused, 488(0) Mapped, 484(4) M swapon, Network: packets: 1807542/4438 in, 5751922/23M out.																											
Disks: 1186378/28G read, 688999/9085M written.																											
PID	COMMAND	NCPU	TIME	ATH	#WG	#PORT	MEM	PURG	CMRPS	PGRP	PPID	STATE	BOOSTS	NCPU	MEM	NCPU_OTHERS	UID	FAULTS	COW	MSGSENT	MSGRCV	SYSSBD	SYMSZ	CSW	PAGEIN	IDLEW	POWE
0	kernel_task	18.0	04:16:18	555/0	0		17M	08	08	0	0	running	[0]	0.00000	0.00000	0	34781+	2	365644+	315488+	0	0	8135423+	0	85178+	0.0	
1865	Comet Helper	18.0	04:16:17	74/21	3	227	110M	08	43M	965	965	sleeping	*[22117]	1.17000	0.00000	501	79634+	933	45080+	19543+	433618+	174279+	51858+	1517	474+	18.0	
965	Comet	0.4	01:18:59	41	1	692	131M	08	90M	965	1	sleeping	*[28315]	0.00000	0.00000	501	131754	1211	457493+	45679+	2488661+	198164+	17252	1938+	8.5		
487	WindowServer	6.3	03:56:57	20	5	4457+	694M	21M+	231M	487	1	sleeping	[011]	0.07162	1.48336	88	318348+	18429	2193377+	1747853+	2498327+	5442262+	1892749+	5556	2286+	5.4	
875	Google Chrom	4.7	03:25:08	19	1	141	41M	08	23M	441	441	sleeping	[013]	0.00000	0.00000	501	68564	833	39083	16419	1785483+	30954	399769+	4497	4	4.7	
1679	top	4.3	00:01:08	1/1	0	29	7984K	08	88	1675	1675	running	[011]	0.00000	0.00000	0	7436+	95	988521+	498272+	36294+	541654+	464+	29	0	4.3	
441	mDNSResponder	4.2	00:28:57	4	2	132	9898K	08	1688K	441	1	sleeping	[011]	0.00000	0.00000	65	14379	154	8073+	4434+	843245+	14867+	192288+	566	1845+	4.4	
558	com.apple.dt	2.7	00:32:06	0	6	1346	30M	08	4192K	558	1	sleeping	[011]	0.00000	0.00000	278	28834+	63	74194+	57716+	29874+	611488+	48836+	1191	4310+	10.1	
874	Google Chrom	2.3	00:41:31	24	5	385	350M	08	48M	441	441	sleeping	[115]	0.05633	0.00000	501	99436+	926	383233+	135948+	294889+	889367+	347713+	3644	287	2.3	
626	trustd	1.8	00:19:28	3	2	259	9648K	512K	3748K	626	1	sleeping	[014]	0.00000	1.78015	501	28993+	122	4513+	4186+	118757+	38635+	51577+	91	1	1.8	
441	Google Chrom	1.7	00:45:48	50	2	958	160M	08	37M	441	1	sleeping	[018]	0.00177	0.00000	501	224688+	1323	356614+	156147+	481568+	1198861+	432739+	27765	62	1.7	
644	WhatsApp	1.6	01:48:41	26	4	688	287M	1024K	135M	644	1	sleeping	[115]	0.18765	0.00000	501	329533+	3292	76418+	26911+	1561529+	628433+	684488+	36979	89	1.6	
399	bluetoothd	1.4	00:08:59	12	6	480	18M	256K	3648K	399	1	sleeping	[011]	1.57497	0.00000	0	25736	184	17639+	7138+	164772+	31535+	73865+	4543	3	1.4	
421	nautilus	1.4	00:03:25	8	5	170	8592K	08	1448K	421	1	sleeping	[014]	2.08168	0.00000	501	14531+	152	5454+	2880+	68381+	11769+	22238+	376	0	1.4	
369	coreduetd	1.2	00:03:19	8	7	98+	7249K	384K	1778K	369	1	sleeping	*[149+]	0.00000	1.16146	0	28888+	158	4626+	2348+	182472+	19378+	3686+	1286	0	1.2	
753	corespotlight	1.1	00:16:16	8	6	382+	25M+	08	14M	753	1	sleeping	[0611]	3.02888	0.00000	501	141911+	381	48392+	26874+	517111+	59183+	282434+	18318	1	1.1	
693	suggestd	0.9	00:07:57	7	6	254+	13M+	08	3712K	693	1	sleeping	[019+]	0.00000	0.08467	501	28713+	243	11841+	2619+	73986+	25498+	57258+	3390	1	0.9	
614	cloudsd	0.7	00:18:55	7	4	297+	16M	08	4848K	614	1	sleeping	[0795+]	0.00000	0.28073	501	42331+	286	47845+	21583+	326218+	85836+	147206+	4551	11	0.7	
978	helpd	0.7	00:03:39	8	6	93	17M+	08	8112K	978	1	sleeping	[014]	0.00000	0.16449	501	6592+	153	1338+	318	44898+	3725+	12675+	993	3	0.7	
788	aid	0.5	00:01:04	2	1	36	2336K	08	488K	788	1	sleeping	[0307+]	0.00000	0.09224	265	7268+	181	3176+	2651+	25778+	11523+	6525+	883	0	0.5	
489	cfprefsd	0.5	00:05:25	3	2	952	3856K	44K	1184K	489	1	sleeping	[01444+]	0.00000	0.05765	0	18482+	98	16584+	16547+	208119+	42144+	48474+	14	0	0.5	
657	Terminal	0.5	00:09:08	9	4	343	151M	21M	14M	657	1	sleeping	[0124+]	0.03113	0.03906	501	73791+	411	85543+	12656+	138892+	288145+	77348+	6365	78	0.5	
653	sharings	0.4	00:08:47	4	2	348	17M	08	8784K	653	1	sleeping	[011]	0.00000	0.13564	501	28686	291	28861+	9211+	98822+	62564+	74175+	3155	1	0.4	
789	Finder	0.4	00:25:28	7	4	549	161M	08	135M	789	1	sleeping	[213]	0.04518	0.00000	501	157778+	542	139782+	58225+	186248+	356963+	155165+	7435	115	0.4	
568	searchpartyd	0.4	00:02:39	5	8	89	8359K	08	3824K	568	1	sleeping	[01903+]	0.00000	0.32918	0	11258	156	1855+	1551+	39188+	4429+	18889+	2483	0	0.4	
681	afirm2nd	0.4	00:15:53	2	72	6881K	72	6881K	72	6881K	681	1	sleeping	*[51121]	0.00000	0.00000	0	7848	165	2498	18544+	45822+	28484+	1312+	298	0	0.4
382	locationd	0.4	00:05:03	7	4	915	9985K	08	3948K	382	1	sleeping	[0685+]	0.00000	0.38328	205	18257	179	88612+	7183+	71526+	18792+	48616+	43214	9	0.4	
334	logd	0.4	00:09:55	4	3	2846	22M	08	29M	334	1	sleeping	[011]	0.00000	0.00000	0	28478+	98	59378+	68588+	329278+	71573+	125414+	66	1	0.4	
618	reportd	0.3	00:02:49	3	287	19M	08	4520K	618	1	sleeping	[011]	0.00000	0.38525	501	13884	217	4377	4832+	36446+	11887+	17057+	1842	1	0.3		
818	netbiosd	0.3	00:01:49	7	6	31	2752K	08	1824K	818	1	sleeping	[011]	0.00000	0.00000	222	5544	155	58	92487+	719	25555+	14871+	867	0	0.3	
542	mysqld	0.3	00:03:04	36	0	58	450M	08	448M	542	1	sleeping	[011]	0.00000	0.00000	74	34838	131	16449	888+	651126+	1484+	92837+	1651	621+	0.3	
749	netatmd	0.3	00:03:57	4	94	4546K	08	129M	749	1	sleeping	[017]	0.00000	0.24595	268	6458	127	6777	1235+	38821+	6423+	13318+	476	0	0.3		
684	aid	0.3	00:01:49	5	3	136+	7128K	08	4808K	684	1	sleeping	[0198+]	0.00000	0.24682	501	11373+	152	4239+	2455+	35164+	11334+	14781+	1860	0	0.3	
761	Logisim-evl	0.2	00:06:58	41	2	374	27M	08	192M	761	1	sleeping	[013]	0.00000	0.00000	501	59773	347	12943	4631	89458+	35687	77675+	7291	289	0.2	
1813	Google Chrom	0.2	00:15:53	31	3	179M	724M	08	179M	641	441	sleeping	[010]	0.00000	0.00000	501	168538	2695	52541+	45783+	71595+	18553+	13564+	5880	190	0.2	
1183	Comet Helper	0.2	00:08:58	17	1	134	26M	08	21M	965	965	sleeping	[0174]	0.00000	0.00000	501	15563	887	13857+	5883+	2639	12588+	7339+	488	1	0.2	
376	asnd	0.2	00:07:53	4	3	398	9873K	08	1924K	376	1	sleeping	[011]	0.00000	0.00000	0	36257+	161	29347+	13494+	488362+	182013+	67688+	1183	0	0.2	
1502	Google Chrom	0.2	00:14:47	24	1	248	331M	08	259M	441	441	sleeping	[010]	0.00000	0.00000	501	258469	1958	91626+	55485+	145107	132785+	145674+	61789	2	0.1	
482	corebrightness	0.2	00:04:06	2	1	126	4112K	08	2128K	482	1	sleeping	[011]	0.15756	0.00000	0	8661+	129	15149+	139279+	33349+	41791+	877	0	0.2		
593	cfprefsd	0.1	00:07:58	3	2	876	4112K	33K	1688K	593	1	sleeping	[01938+]	0.00000	0.15215	501	41552+	91	21346+	28095+	282112+	54228+	67189	2	0	0.1	
775	spotlightktr	0.1	00:02:44	4	274	7745K	08	2644K	775	1	sleeping	[0197]	0.00000	0.00000	501	12853+	178	3584+	1983+	68826+	7614+	34853+	1154	0	0.1		
363	mds	0.1	00:01:59	8	5	328	33M	08	89M	363	1	sleeping	[011]	0.01277	0.00000	0	35462+	78	48553+	67471+	645978+	788739+	146888+	1581	3	0.1	
512	audiodcheckd	0.1	00:08:77	4	3	58	2736K	08	1397K	512	1	sleeping	[011]	0.00000	0.18978	0	9891	138	8466+	7562+	78999+	18274+	22549+	1856	3	0.1	
749	reactivity	0.1	00:03:57	1	122	36M	36M	08	1854K	749	1	sleeping	[0133+]	0.00000	0.11255	501	14633+	178	3584+	1983+	68826+	7614+	34853+	1154	0	0.1	
338	fsaverd	0.0	00:07:08	12	1	176	4272K	08	1184K	338	1	sleeping	[011]	0.00000	0.00000	0	18916	108	17482+	10287	131988+	11886+	13491+	1268	41	0.0	
348	powerd	0.0	00:01:56	3	2	153	4992K	08	1369K	348	1	sleeping	[011]	0.00000	0.06459	12884+	120	18378+	18749+	4511+	29815+	22252+	260	0	0.0		
432	Comet Helper	0.0	00:01:56	3	2	153	4992K	08	1369K	348	1	sleeping	[011]	0.00000	0.06459	12884+	120	18378+	18749+	4511+	29815+	22252+	260	0	0.0		
432	Comet Helper	0.0	00:01:56	3	2	153	4992K	08	1369K	348	1	sleeping	[011]	0.00000	0.06459	12884+	120	18378+	18749+	4511+	29815+	22252+	260	0	0.0		
432	Comet Helper	0.0	00:01:56	3	2	153	4992K	08	1369K	348	1	sleeping	[011]	0.00000	0.06459	12884+	120	18378+	18749+	4511+	29815+	22252+	260	0	0.0		
432	Comet Helper	0.0	00:01:56	3	2	153	4992K	08	1369K	348	1	sleeping	[011]	0.00000	0.06459	12884+	120	18378+	18749+								

Command	Resource/Information	OS Concept
uname -a	Kernel and system information	Kernel abstraction
lscpu	CPU architecture and cores	CPU management
ps aux	Running processes	Process management
top	Real-time CPU/memory/process usage	Resource monitoring

Conclusion

The experiment demonstrates that the **operating system acts as an abstraction layer between applications and hardware**. The `uname` command provides information about the Linux kernel and system architecture, while `lscpu` reveals CPU resources managed by the OS. The `ps` command shows how the OS manages individual processes, and `top` provides real-time information about CPU, memory, and process utilization.

The OS abstracts **CPU through process scheduling, memory through virtual memory, storage through files and file systems, and I/O devices through device drivers and system calls**. Therefore, applications can use hardware resources without directly communicating with the physical hardware. This abstraction improves **security, resource sharing, portability, and efficient system management**.